

Joanna Piper Morgan, PhD

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RESEARCH EXPERIENCE

Lawrence Livermore National Lab, Nuclear Criticality Safety Division September 2025
Postdoctoral Research Scholar

Subject: TEX Experiment design and evaluation, Sn and Monte Carlo algorithms development using MFEM and PETSc.

Oregon State University, School of MIME June 2020 – June 2025
Graduate Research Assistant

Subject (1): Python based acceleration and abstraction of compute kernels for dynamic Monte Carlo in a rapid methods development code MC/DC as part of the Center for Exascale Monte Carlo Neutron Transport (CEMeNT)

Subject (2): Investigation of the one cell inversion method as an alternative to transport sweeps for deterministic dynamic neutron transport on GPUs

Mentors: Kyle E. Niemeyer & Todd S. Palmer

Advanced Micro Devices (AMD), HPC GPU Apps Support Group September 2023 – May 2024

Co-Op Research Intern

Subject: Python–Numba compiler bindings for AMD GPUs and Sn numerical methods in roc-SOLVER, profiler testing for MI300a

Mentors: Damon McDougall, Christopher Kime

Los Alamos National Laboratory, XCP-3 June 2022 – May 2023
Graduate Research Intern

Subject: Implementing Woodcock delta tracking on a structured mesh within the production code MCATK. This work was pushed to a production version release.

Mentors: Travis J. Trehan, Timothy P. Burke, & Collin J. Josey

Los Alamos National Laboratory, CCS-2 June 2021 – June 2022
Graduate Research Intern

Subject: Novel methods exploration in vectorizable variance reduction for thermal radiation transport (TRT)

Mentors: Kendra Long & Alex Long

Argonne National Laboratory June 2019 – September 2019
Lee Teng Undergraduate Research Fellow

Subject: Cancer radiotherapy scanner magnet design & analysis

Mentor: Brahim Mustapha

Thomas Jefferson National Accelerator Facility June 2018 – August 2018
Science Undergraduate Laboratory Intern (SULI)

Subject: Superconducting quadrupole magnet optimization

Mentor: Renuka Rajput-Ghoshal

EDUCATION

Oregon State University (OSU), Corvallis, OR September 2020 – June 2025
Ph.D., Mechanical Engineering

- **Minor:** Nuclear Engineering, **Emphasis:** Thermal Fluid Sciences (TFS)
- **Dissertation Title:** Algorithms and Software Engineering Schemes for Time-Dependent Radiation Transport on Heterogeneous Compute Architectures.
- **Academic Advisor:** Kyle E. Niemeyer, **Minor Advisor:** Todd S. Palmer

Oregon State University (OSU), Corvallis, OR

September 2020 – March 2022

M.S., Mechanical Engineering

- **Emphasis:** Thermal Fluid Sciences (TFS), **GPA:** 3.83
- **Project Report:** Explorations of Monte Carlo Solution and Implementation Methods for Thermal Radiation and Neutron Transport
- **Academic Advisor:** Kyle E. Niemeyer

Oregon Institute of Technology (OIT), Klamath Falls, OR

September 2016 – June 2020

B.S., Mechanical Engineering, magna cum laude

- **Minors:** Applied Physics; Applied Mathematics

JOURNAL PUBLICATIONS

1. J. P. Morgan, I. Variansyah, K. B. Clements, T. S. Palmer, & Niemeyer, K. E. (2026). Hybrid Delta Tracking Schemes Using a Track-Length Estimator. *Journal of Computational and Theoretical Transport*, 1–33. [doi:10.1080/23324309.2026.2618791](https://doi.org/10.1080/23324309.2026.2618791) arxiv:2510.00152.
2. J. P. Morgan, I. Variansyah, T. S. Palmer, and K. E. Niemeyer. (2025) One-Cell Inversion for Solving Higher-Order Time-Dependent Radiation Transport on GPUs. Accepted. *Nuclear Science and Engineering*. 200(5) 1074-1093 [doi:10.1080/00295639.2025.2510004](https://doi.org/10.1080/00295639.2025.2510004). arXiv:2503.00264.
3. J. P. Morgan, I. Variansyah, B. Cuneo, T. S. Palmer, and K. E. Niemeyer. (2025) Performant and Portable Monte Carlo Neutron Transport via Numba. *Computing in Science and Engineering (IEEE)* 27(1), pp. 57-65. [doi:10.1109/MCSE.2025.3550863](https://doi.org/10.1109/MCSE.2025.3550863). arXiv:2409.04668.
4. J. P. Morgan, A. Mote, S. Pasmann, G. Ridley, T. S. Palmer, K. E. Niemeyer, R. G. McClarren. (2024) “The Monte Carlo Computational Summit – October 25 & 26, 2023 – Notre Dame, Indiana, USA”. *Journal of Computational and Theoretical Transport*. 53 (5), 361–382. [doi:10.21105/joss.06415](https://doi.org/10.21105/joss.06415).
5. J. P. Morgan, I. Variansyah, S. Pasmann, K. B. Clements, B. Cuneo, A. Mote, C. Goodman, C. Shaw, J. Northrop, R. Pankaj, E. Lame, B. Whewell, R. McClarren, T. Palmer, L. Chen, D. Anistratov, C. T. Kelley, C. Palmer, and K. E. Niemeyer. (2024) “Monte Carlo / Dynamic Code (MC/DC): An accelerated Python package for fully transient neutron transport and rapid methods development”. *Journal of Open Source Software*. 9(96), 6415. [doi:10.1080/23324309.2024.2354401](https://doi.org/10.1080/23324309.2024.2354401). arXiv:2402.08161.

CONFERENCE PUBLICATIONS

1. J. P. Morgan, J. Glesmann, C. Percher, Z. Lemke, Experiment Completion: Thermal Epithermal eXperiments (TEX) with Iron Diluents for Hanford Tank Safety Basis. (2026) *Transactions of the American Nuclear Society*, Phoenix, AZ.
2. J. P. Morgan. Second Moment Acceleration for Incident Fluxes on Cell Surfaces. (2026) *29th International Conference on Transport Theory (ICTT-29)*.
3. J. P. Morgan, C. Percher, E. Aboud. Revision Two and Three Comparison from Flattop HEU Benchmark (HEU-MET-FAST-028). (2026) *Transactions of the American Nuclear Society*, Denver, CO.

4. J. P. Morgan, B. Cuneo, I. Variansyah, K. E. Niemeyer. (2025) “Enabling GPU portability into the Numba-JITed Monte Carlo particle transport code MC/DC”. *Proceedings of the International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering (ANS M&C 2025)*. Denver, CO, USA. doi:10.13182/MC25-47142. arXiv:2501.05440.
5. B Cuneo, J. P. Morgan, I. Variansyah, K. E. Niemeyer. (2025) “Comparing the Performance of MC/DC’s on-GPU Event-based Processing Methods in Multigroup and Continuous-energy Problems”. *Proceedings of the International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering (ANS M&C 2025)*. Denver, CO, USA. doi:10.13182/MC25-47174. arXiv:2506.00263.
6. J. P. Morgan, I. Variansyah, T. S. Palmer, and K. E. Niemeyer. (2023) “Exploring One-Cell Inversion Method for Transient Transport on GPU.” *Proceedings of the International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering*. Niagara Falls, Ontario, Canada. arXiv:2305.13555.
7. J. P. Morgan, T. J. Trahan, T. P. Burke, C. J. Josey, and K. E. Niemeyer. (2023) “Hybrid-Delta Tracking on a Structured Mesh in MCATK.” *Proceedings of the International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering*. Niagara Falls, Ontario, Canada. arXiv:2306.07847.
8. I. Variansyah, J. P. Morgan, K. E. Niemeyer, and R. G. McClarren. (2023) “Development of MC/DC: a performant, scalable, and portable Python-based Monte Carlo neutron transport code.” *Proceedings International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering*. Niagara Falls, Ontario, Canada. arXiv:2305.07636.
9. J. P. Morgan, A. Long, K. Long, and K. E. Niemeyer. (2022) “Novel MC TRT Method: Vectorizable Variance Reduction for Energy Spectra” In *Transactions of the American Nuclear Society*, volume 126, pp. 276–278. Anaheim, California, USA. doi:10.13182/T126-38137. zenodo:6646813.
10. J. P. Morgan, T. S. Palmer, and K. E. Niemeyer. (2022) “Explorations of Python-Based Automatic Hardware Code Generation for Neutron Transport Applications.” In *Transactions of the American Nuclear Society*, volume 126, pp. 318–320. Anaheim, California, USA. doi:10.13182/T126-38066. zenodo:6643659.

PUBLICATIONS IN PREPARATION

1. J. P. Morgan, T. S. Palmer, and K. E. Niemeyer. Efficient Preconditioning for Space-Parallel One Cell Inversions in Slab Geometry using a Second Moment Method. *in preparation*
2. M. Derman, J. P. Morgan, K. E. Niemeyer, T. S. Palmer. Unnamed paper on complex eigenvalue decomposition. *in preparation*

CONFERENCE AND POSTER PRESENTATIONS

- J. P. Morgan, June 2026
- J. P. Morgan, C. Percher, E. Aboud. May 2026
- J. P. Morgan, B. Cuneo, I. Variansyah, K. E. Niemeyer. May 2025. Enabling GPU Portability into the Numba-JITed Monte Carlo Particle Transport Code MC/DC. International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering (M&C 2025), Denver, CO

- J. P. Morgan. February 2025. AMD Tools Development Workshop, University of Oregon, Eugene, OR
- J. P. Morgan, K. E. Niemeyer. July 2024. Performant and Portable High-Performance Computing at Scale via Python and Numba. SciPy. Tacoma, WA
- J. P. Morgan. “Exploring One-Cell Inversions for Transient Transport on GPUs.” Copper Mountain Conference on Iterative Methods (SIAM). April 2024.
- J. P. Morgan. “High Performance Python for Rapid Methods Development in Monte Carlo / Dynamic Code.” Sustainable Scientific Software Conference. April 2024.
- J. P. Morgan, I. Variansyah, T. S. Palmer, and K. E. Niemeyer. “Exploring One-Cell Inversion Method for Transient Transport on GPU.” High Energy Density Summer School, San Diego, California. July 2023
- J. P. Morgan, I. Variansyah, T. S. Palmer, and K. E. Niemeyer. “Exploring One-Cell Inversion Method for Transient Transport on GPU.” International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering. Niagara Falls, Ontario, Canada (2023).
- J. P. Morgan, T. J. Trahan, T. P. Burke, C. J. Josey, and K. E. Niemeyer. “Hybrid-Delta Tracking on a Structured Mesh in MCATK.” International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering. Niagara Falls, Ontario, Canada (2023).
- J. P. Morgan, T. S. Palmer, & K. E. Niemeyer, “Hardware Code Generation Techniques for Accelerating Python.”, Scientific Python, Austin, Texas, USA, July 2022
- J. P. Morgan, T. S. Palmer, & K. E. Niemeyer, “Hardware Code Generation Techniques for Accelerating Python”, Annual Meeting of the American Nuclear Society, Anaheim, California, USA, June 2022
- J. P. Morgan, A. Long, K. Long & K. E. Niemeyer, “A novel MC TRT method: vectorizable variance reduction for the energy spectra”, Annual Meeting of the American Nuclear Society, Anaheim, California, USA, June 2022
- J. P. Morgan and B. Mustapha, “Carbon therapy X-Y scanner magnet analysis,” in Lee Teng Internship - Posters and Final Report Presentations, Fermi National Accelerator Facility, Batavia, Illinois. August 2019.
- J. P. Morgan and R. Rajput-Ghoshal, “Jefferson lab electron ion collider interaction region quadrupole magnet optimization,” in Undergraduate Research Poster Presentations, Thomas Jefferson National Accelerator Facility, Newport News, Virginia, August, 2018.

CONFERENCE ATTENDANCE AND PROFESSIONAL DEVELOPMENT

** presented at*

- November 2026, 2026 ANS Winter Conference & Expo
- June 2026, 29th International Conference on Transport Theory (ICTT-29)*, Aix-en-Provence, France

- June 2026, American Nuclear Society Annual Conference*, Denver, CO
- April 2026, International Criticality Safety Benchmark Evaluation Project (ICSBEP) Technical Review Meeting, Paris, France
- January 2026, Nuclear Data Week (Cross Section Evaluation Working Group (CSEWG)*, Brookhaven National Lab, NY
- December 2025, US Research Software Sustainability Institute (URSSI)*, Portland, OR
- May 2025, International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering (M&C 2025), Denver, CO*
- February 2025, AMD Tools Development Workshop*, University of Oregon, Eugene, OR
- July 2024, Scientific Computing in Python (SciPy), Tacoma, WA*
- May 2024, NEA Workshop for Radiation Transport Simulation Developers (RTS 2024), Frascati, Italy
- April 2024, 18th Copper Mountain Conference on Iterative Methods (SIAM), Copper Mountain, CO*
- April 2024, Sustainable Scientific Software Conference (S3C), Seattle, WA*
- January 2024, NUWEST, Albuquerque, NM (unable to attend due to weather)
- January 2024, US Research Software Sustainability Institute (URSSI), Portland, OR. Hosted at Oregon State University Portland Center. Attended as a teaching assistant
- August 2023, International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering (ANS M&C International), Niagara Falls, Ontario, Canada*
- July 2023, High Energy Density Science (HEDS) Summer School, San Diego, CA at University of California San Diego
- July 2022, Scientific Computing in Python (SciPy), Austin, TX*
- June 2022, American Nuclear Society Annual Conference, Anaheim, CA*
- October 2021, Conference on Mathematics and Computational Methods Applied to Nuclear Science (ANS M&C), Raleigh, North Carolina (virtual)
- June 2018, United States Particle Accelerator School (USPAS), Albuquerque, NM. Fundamentals of Accelerator Physics and Technology with Simulations and Measurements Lab

TEACHING EXPERIENCE

Oregon Institute of Technology, Dept. Natural Sciences September 2017 – March 2020
Supplemental Instructor

- **Class:** Physics for medical imaging
- **Directed by:** Robyn Wilde

Oregon Institute of Technology, Student Services January 2018 – June 2020
Peer Consultant

- **Subjects:** Numerical methods, differential & integral calculus, linear algebra, intro to algebra, trigonometry, college algebra, statistics, heat transfer, statics, thermodynamics, fluid dynamics, strengths of materials, physics with calculus.

Oregon Institute of Technology, Dept. Mathematics

January 2018 – March 2020

Mathematics Grader

- **Subjects:** Numerical methods (in MATLAB), differential & integral calculus, linear algebra, intro to algebra, trigonometry, college algebra, statistics.
- **Directed by:** Cristina Negoita and Terri Torres

PROFESSIONAL AFFILIATIONS

- Member, American Nuclear Society (ANS)
- Member, Out in Science Technology Engineering and Mathematics (oSTEM)
- Member, Tau Beta Pi (Engineering Honors Society), Oregon Delta 2018

OTHER ACTIVITIES

- Volunteer; Whiteside Theater, Corvallis, OR (March 2023 – June 2025)
- Student Commissioner; Library Recourse Commission (October 2017 – June 2020)
- Committee Member; University Librarian Search Committee (March 2018 – June 2018)
- Student Building Manager; Oregon Institute of Technology College Union (February 2017 – March 2020)
- President; Oregon Institute of Technology Chapter of Circle K International Community Service Club (May 2019 – June 2020)
- Treasurer; Oregon Delta – Tau Beta Pi (May 2019 – May 2020)
- Notary Public; State of Oregon (November 17th, 2016 – November 16th, 2020)

AWARDS AND FELLOWSHIPS

- Illinois Accelerator Institute; Lee Teng Undergraduate Research Fellowship in Accelerator Science (2019)
- ASME; Irma and Robert Bennett Scholarship (2019)
- Pride Foundation; Lenehan-Warn Technical Education Scholarship (2017)

SKILLS

- **Clearance:** Active Q security clearance
- **Languages:** PETSc (beginner-intermediate), MFEM (beginner-intermediate), C/C++ (OpenMP, C-CUDA, HIPC C/ROCm, Kokkos, AVX / SIMD, MPI, Intel MKL, LAPACK) (beginner-intermediate), Python (Numba, mpi4py, Numpy, Scipy, Matplotlib, CuPy, CProfiler) (intermediate), Matlab (intermediate), VBA in Excel, FORTRAN (beginner), LLVM (beginner-intermediate)
- **Compilers:** Numba (LLVM bindings for Python), Intel Compilers, HIP and ROCm (Clang) compilers, GCC, CRAY Compilers
- **Software Development Tools:** Profilers (Intel Vtune, NVIDIA N-Sight, TotalView), Debuggers (valgrind, gdb)
- **Engineering Design and Simulation Tools:** CAD/CAE (Creo, SolidWorks, AutoCAD, Inventor, NX, CST Studio Suite, Opera, FEMM)

- **Nuclear Engineering Codes:** MCNP, MCATK (developed in), MC/DC (developed in), Shift (developed in)
- **Visualization Toolkits:** Paraview, Visit
- **MISC Development Tools:** Services (Git(hub), Gitlab, Bitbucket), Terminal (Bash, z-shell, ssh & X-11 forwarding), Package Publishing (Conda, PyPi, Sphinx, Read the Docs), Continuous Integration (Github actions, Jakamar runners), Website Builders (Sphinx, Read the Docs, Jekyll, Ruby), LaTeX

REFERENCES

Available upon request